

Polynomials: Combine and Distribute

Combine Like Terms

$$ax^n$$

Diagram labels: Coefficient (points to a), Variable(s) (points to x), Exponent (points to n)

'Like' terms must have

- same variable(s)
- same exponent(s)

Combine by

- adding/subtracting coefficients of like terms

Example:

$$3x^2y^3 + 10xy - 5x^2y^3 - 20xy + 4 = -2x^2y^3 - 10xy + 4$$

Diagram shows grouping like terms: $3x^2y^3$ and $-5x^2y^3$ are grouped together, as are $10xy$ and $-20xy$.

Ask: How Many?

x^2y^3	xy	x^0 (constants)
$3x^2y^3 - 5x^2y^3 = -2x^2y^3$	$+10xy - 20xy = -10xy$	$+4$

Distributive Property

$$a(b + c) = ab + ac$$

- Multiply each term in the second polynomial by every term in the first polynomial
- Combine like terms to simplify

Monomials: $a(b + c + d + e) = ab + ac + ad + ae$

Diagram shows arrows from a to each term inside the parentheses.

Example:

$$-3x(7x^2 + 4x - 12) = -21x^3 - 12x^2 + 36x$$

Polynomials: $(a + b)(c + d) = ac + ad + bc + bd$

Diagram shows arrows from each term in the first polynomial to each term in the second polynomial.

Examples:

$$(2x - 3)(5x - 8) = 10x^2 - 16x - 15x + 24$$

$$= 10x^2 - 31x + 24$$

$$(x + y)(2x - 3y + z) = 2x^2 - 3xy + xz + 2xy - 3y^2 + yz$$

$$= 2x^2 - xy + xz - 3y^2 + yz$$

